

School of Arts and Sciences

Department of Computer Science and Information Technology

Fall 2025-2026

CSCI300 Intermediate Programming with Objects

Class Day : _____ **Class Time:** _____ **Room:** _____
Instructor : _____
E-mail : _____
Office : _____ **Office Hours:** _____ **Extension No:** _____

I. Course Description

The course emphasizes the principles of object-oriented programming using the Java Programming Language. It starts by an introduction to creating applications using Java. Then, the course introduces how to define classes and declare objects and discusses the main topics related to Object Oriented Programming (constructors, methods, dependency, aggregation, inheritance, and polymorphism). Finally, the course introduces exception handling as well as writing to and reading from files.

II. Course Objectives

CO-1: To learn various object oriented programming concepts and properties: objects, classes, private/public members, constructors, encapsulation, inheritance and polymorphism.

CO-2: To learn the advantages of programming in an object-oriented language such as Java.

III. Course Learning Outcomes

CLO-1: Explain and use the principles of object-oriented programming including encapsulation, composition, inheritance, and polymorphism.

CLO-2: Demonstrate good skills in creating and using classes, methods, objects, and UML class diagrams.

CLO-3: Distinguish the difference between concrete classes, abstract classes, and interfaces.

CLO-4: Show the ability to work with input and output text files.

CLO-5: Apply exception handling techniques to deal with exceptions occurrence

IV. Mapping to program outcomes (*PLOs are listed at the end of the syllabus*)

Intermediate Programming with Objects	PLO -1	PLO -2	PLO -3	PLO -4	PLO -5	PLO -6	PLO -7	PLO -8	PLO -9
CLO-1	X								
CLO-2		X	X						
CLO-3	X								
CLO-4		X	X						
CLO-5	X		X						

V. Topics Covered

1. Object oriented concepts and properties
2. Classes, private/public members
3. Encapsulation, inheritance, and polymorphism
4. Exception handling
5. Files

VI. Prerequisites – Corequisites

Prerequisite	Corequisite
CSCI250 Introduction to Programming	CSCI300L Intermediate Programming with Objects Lab

VII. Course Material:

Textbook Introduction to Java Programming, Daniel Liang, 9th edition

Publisher: Pearson; **ISBN-10:** 0132936526

VIII. Software(s) Required

JDK and NetBeans

IX. Course Assessment

# Activity	Percentage
Quizzes	10%
Test	20%
Midterm	35%
Final Exam	35%

I. Detailed Schedule

Week		Book Reference	Topic Details
Week 1 Sep 29 - Oct 2	Lecture 1	Introduction to Java Programming	Syllabus: reading and explaining. Explain what Object Oriented Programming (OOP) is. Give a brief overview of objects, attributes, and behaviors. Discuss the relation/difference between Structured Programming taught in Introduction to Programming (CSCI250) and OOP taught in Intermediate Programming with Objects (CSCI300). 1.1 Review of chapter 4 Loops and chapter 5 Methods from Introduction to Programming (CSCI250).
	Lecture 2	Introduction to Java Programming	Review of chapter 5 Methods and chapter 6 Single dimensional Arrays from CSCI250.
Week 2 Oct 6 - 9	Lecture 1	Introduction to Java Programming	Chapter 8 Objects and Classes 8.2 Defining Classes for Objects: Emphasis on the state of an object (Attributes), behavior (Methods) The Circle class example with emphasis on attributes (data fields), behaviors (Methods), and the UML class diagram 8.3 Example: Defining classes and Creating Objects (with emphasis on UML Diagrams) The Circle class and an application (program) to test the Circle class 8.4 Constructing Objects Using Constructors.
	Lecture 2	Introduction to Java Programming	Solve a modified version (will be provided then) of programming exercise 8.7 The Account class. 8.5 Accessing Objects via Reference Variables. 8.5.1 Reference Variables and Reference Types 8.5.2 <i>Accessing</i> an Object's Data and Methods 8.5.3 Reference Data Fields and the null Value. 8.5.4 Differences between Variables of Primitive Types and reference Types.
Week 3 Oct 13- 16	Lecture 1	Introduction to Java Programming	8.6 Using Classes from the Java Library 8.6.1 The Date Class 8.6.2 The Random Class 8.7 Static Variables, Constants, and Methods
	Lecture 2	Introduction to Java Programming	8.8 Visibility Modifiers (public, Private). 8.9 Data Field Encapsulation (set and get Methods).

Week 4 Oct 20 - 23	Lecture 1	Introduction to Java Programming	Explain the toString() method and the print() method using the example of the Circle class 8.10 Passing Objects to Methods 8.11 Array of Objects.
	Lecture 2	Introduction to Java Programming	Quiz 1- Chapter 8 (output) Solve extra problem on chapter 8
Week 5 Oct 27 - 30	Lecture 1	Introduction to Java Programming	Assignment 1 (Chapters 8 & 9) due on Sunday Chapter 9 Strings and Text I/O 9.2 The String Class 9.2.1 Constructing a String 9.2.3 String Comparisons 9.2.4 String Length, Characters, and Combining Strings 9.2.5 Obtaining Substrings 9.2.6 Converting, Replacing, and Splitting Strings
	Lecture 2	Introduction to Java Programming	.2.8 Finding a Character or a Substring in a String 9.2.10 Converting Characters and Numeric Values to Strings Converting Strings to Integer and Double using Integer.parseInt() and Double.parseDouble() Solve programming exercises 9.2 and 9.4
Week 6 Nov 3 - 6	Lecture 1	Introduction to Java Programming	NON COMMON TEST Chapter 8
	Lecture 2	Introduction to Java Programming	Chapter 10 Thinking in Objects 10.3 The Scope of Variables 10.4 The this Reference 10.7 Object Composition 10.8 Designing the course Class

Week 7 Nov 10-13	Lecture 1	Introduction to Java Programming	Assignment 2 (Chapter 10) due on Sunday 10.5 Class Abstraction and Encapsulation 10.6 Object-Oriented Thinking Solve programming exercises 10.4 and 10.12
	Lecture 2	Introduction to Java Programming	Quiz 2 – Chapter 9(output) Chapter 11 Inheritance and Polymorphism 11.2 Super classes and Subclasses Listings (Examples) 11.1, 11.2, 11.3, 11.4
Week 8 Nov 17- 20	Lecture 1	Introduction to Java Programming	11.3 Using the super Keyword 11.3.1 Calling Superclass Constructors 11.3.2 Constructor Chaining 11.3.3 Calling Superclass Methods
	Lecture 2	Introduction to Java Programming	11.4 Overriding Methods 11.5 Overriding vs. Overloading 11.6 The Object Class and its toString() Method
Week 9 Nov 24-27	Lecture 1	Introduction to Java Programming	11.7 Polymorphism 11.8 Dynamic Binding 11.9 Casting Objects and the instanceof Operator 11.10 The Object's equals Method
	Lecture 2	Introduction to Java Programming	11.11 The ArrayList Class (difference between ArrayList and ArrayList<String>) 11.13 The Protected Data and Methods 11.14 Preventing Extending and Overriding
Week 10 Dec 1-4	Lecture 1	Introduction to Java Programming	Assignment 3 (Chapter 11) due on Sunday Solve programming exercises among 11.2, 11.3, 11.5 and 11.6
	Lecture 2	Introduction to Java Programming	Chapter 15 Abstract Classes and Interfaces 15.2 Abstract Classes 15.2.1 Why Abstract Methods 15.2.2 Interesting Points on Abstract Classes

Week 11 Dec 8-11	Lecture 1	Introduction to Java Programming	Quiz 3 – Chapter 11(ArrayList output) Solve programming exercises 15.2 and 15.8
	Lecture 2	Introduction to Java Programming	15.5 Interfaces 15.6 The Comparable Interface 15.8 Interfaces vs. Abstract Classes Solve programming exercise 15.12
Week 12 Dec 15-18	Lecture 1	Introduction to Java Programming	Chapter 14 Exception Handling and Text I/O 14.1 Introduction 14.2 Exception Handling Overview 14.3 Exception Types (ArithmeticException, NullPointerException, IndexOutOfBoundsException)
	Lecture 2	Introduction to Java Programming	14.3 Exception Types (InputMismatchException, IllegalArgumentException, Exception) 14.4 More on Exception Handling 14.5 The finally Clause 14.9 Creating Custom Exception Class Solve programming exercise 14.5
Week 13 Dec 22, 23	Lecture 1	Introduction to Java Programming	14.10 The File Class 14.11 File Input and Output 14.11.1 Writing Data Using PrintWriter IOException 14.11.2 Reading Data Using Scanner IOException and FileNotFoundException
Week 14 Jan 5 - 8	Lecture 1	Introduction to Java Programming	Tuesday 6 January off (Armenian Christmas) Assignment 4 (Chapter 15 and Chapter 14 – Exceptions) due on Sunday Solve programming exercises 14.11 and 14.13
	Lecture 2	Introduction to Java Programming	Catch up session- preparation for the final exam

I. Class Rules & Discipline

Attendance Policy

The student is administratively withdrawn (AW) from the class if he/she is absent for more than one third of the classes (15 class sessions for courses offered 3 times a week, 10 for courses offered twice a week, and 5 for courses offered once a week). If the number of absences exceeds the limit after the withdrawal deadline, i.e. including all the period up-to the end of the semester, the student will not receive an AW grade, therefore, the final examination must be written, otherwise, an F grade will be granted. When absent, you are responsible for all material presented in class. Missed exams will only be excused by documented evidence as explained in the make-up examination policy below.

A "10%" has been allocated on the "Marking Profile" for attendance. A student who hasn't missed any lecture will get a full mark on the attendance grade. Every absence, a 20% will be deducted, that is, a student with 5 absences will get a ZERO on that grade.

Cheating

The University policy on cheating will be strictly applied. Students are required to adhere to the policy on academic irregularities contained in the LIU student handbook. Collaboration among students in solving assignments is not acceptable. It is okay to ask another student for advice when stuck on a problem, but if the instructor receives assignments that are substantially equivalent, all involved students will be punished with a grade of zero.

Classroom Manners

All students are expected to conduct themselves in a professional manner. Students should not: keep their cell phones on during the class, disrupt the class, do other course work in class, have side conversations with other students, be rude towards anyone. The first failure to respect these rules will result in a *warning* from the instructor. If a student chooses to disregard the *warning*, he/she will be dismissed from class.

Instructor late to the class

In case the instructor is late the students should wait for 15 minutes before leaving the class.

Abuse of Computing Privileges

The Office of Information Technology maintains policies posted in the computer labs regarding proper behavior in the labs. Failure to adhere to these policies may result in loss of computer privileges, and possible legal action.

Make-Up Examination Policy

As per university council decision, a student is eligible for a mid-term or final examination make-up if and only if he/she had the following incidents:

- a. Sickness; **proved by hospitalization report**; that is; a discharge summary is necessary.
- b. Death in the family **proved by a death certificate** and personal identification.
- c. Accidents **proved by an expert report**.

A student that misses an exam should submit a petition within one week from the examination date. Late petitions will be rejected unless it is an extreme case (Hospitalization as an example).

In case a petition is accepted, the student submits a MAKE-UP exam. The grade of the missing exam will be equal to a percentage % of the Final Exam's grade. The percentage % will be set by the administration once the petition is approved.

Cheating policy: Any cheating will be rewarded with a zero, all students involved in cheating will get a zero (even if the cheating is in one question). The activity would not be repeated.

Missing Assignments and Projects Policy

- A -10 per day penalty will be applied to late projects (weekends count as two days).
- A ZERO will be applied for late assignments on the next day.

Curriculum, Yearly Plan, and Advising

If a student majoring in Computer Science or Information Technology is planning to graduate in three years then he/she should follow the corresponding major yearly plan. A copy of the yearly plan and the curriculum (which contains the prerequisites and co-requisites of courses) can be picked up from the office of the school of Arts and Science. When not sure about anything please contact your advisor prior to or during the registration period.

Probation Policy

If a student who has attempted 19 credits has a GPA or a major GPA below 2.0 then the student is considered under probation.

LIU E-mails

LIU students should use their LIU e-mails in communicating with the instructors. Important announcements, syllabi, assignments, and solutions will be sent to your LIU e-mail accounts.

University Peer Review Policy

This is a new policy at LIU in which during the semester, all classes will be visited by instructors either from the same or different departments. This policy helps instructors figure out ways to improve student educational experiences in class and eventually their performance.

Snapshots

Students are not allowed to take snapshots of the whiteboard in classrooms using their mobile-phones.

Equivalent Lettering Grades

<u>Percent (%)</u>	<u>QPTS</u>	<u>Letter Grade (G)</u>	<u>Percent (%)</u>	<u>QPTS</u>	<u>Letter Grade (G)</u>
<60	0	F	67	1.7	C
60	1	D	68	1.8	
61	1.1		69	1.9	
62	1.2		70	2	
63	1.3		71	2.1	
64	1.4		72	2.2	
65	1.5	D+	73	2.3	
66	1.6		74	2.4	
75	2.5	C+	83	3.3	B+
76	2.6		84	3.4	
77	2.7		85	3.5	
78	2.8	B	86	3.6	
79	2.9		87	3.7	
80	3		88	3.8	
81	3.1		89	3.9	
82	3.2		>=90	4	A

<u>Grade</u>	<u>Description</u>	<u>QPTS</u>
A	Excellent	4.00
B+	Very Good	3.50
B	Good	3.00
C+	Above Average	2.50
C	Average	2.00
D+	Fair	1.50
D	Poor	1.00
F	Fail	0.00
IP	Course in Progress	
I	Incomplete work	
W	Student has withdrawn from the course	
AW	Administrative withdrawal from the course	
R	Course has been repeated	
S	Satisfactory work	
U	Unsatisfactory work	
%G	Final Grade	
QPTS	Quality Points	

Computer Science Outcomes

Graduates of the Computer Science Program will have the ability to:

Program outcomes	Program Outcomes
category -1 Underlying Conceptual Basis for Informatics	PLO -1 Describe and explain the essential facts, concepts, theories and mathematical methods relevant to computing, computing equipment, computer communication and informatics applications as appropriate to their program of study
	<i>PLO-1 for CSIT=> Gain basic skills in mathematics and outline the characteristics of relevant state-of-the-art hardware and software and their practical application</i>
category -2 Analysis	PLO -2 Select and use relevant analytic, modelling and simulation methods
	PLO-3 Choose appropriate solution patterns, algorithms and data structures
category -3 Design and Implementation	PLO -4 Describe the software development process and apply relevant practical and programming skills to the creation of computer programs, and/or other informatics artefacts
	<i>PLO-4 for CSIT=>Specify and design computing/network hardware/software which meet specified requirements</i>
category -4 Economic, legal, social, ethical and environmental context	PLO -5 Explain the importance of information privacy and security issues in relation to the design, development, maintenance, monitoring and use of informatics-based systems
category -5 Informatics practice	PLO -6 Describe and explain management techniques appropriate to the design, implementation, testing, deployment and maintenance of informatics systems, including project management, configuration management, change management, etc., and including relevant automated techniques
	<i>PLO -6 for CSIT=> Demonstrate an awareness of project management and business practices, such as risk and change management, and understand their limitations;</i>
category -6 Other Professional Skills and Competences	PLO -7 Communicate effectively both verbally and using a variety of communications media
	PLO -8 Plan self-learning and improve personal performance as a foundation for lifelong learning and ongoing professional development
	PLO -9 Participate effectively in informatics group-working